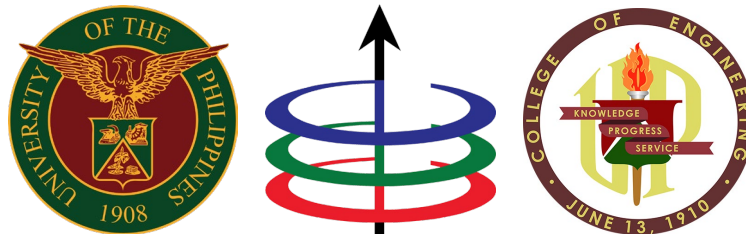


University of the Philippines Diliman
College of Engineering
Electrical and Electronics Engineering Institute

Second-Order Circuits

Natural Response

1st Semester, A.Y. 2026-2027



Student Outcomes

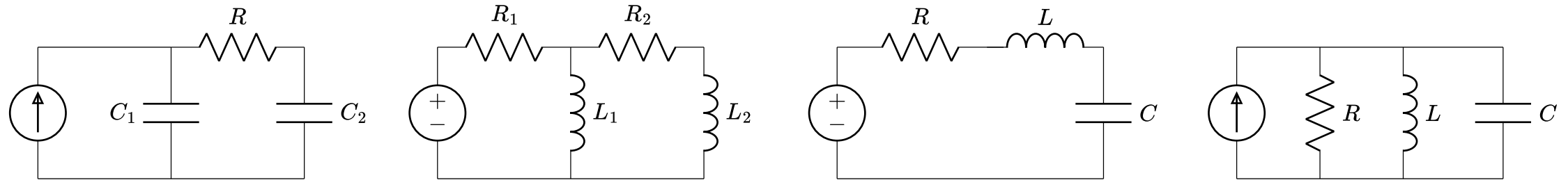
- Determine the natural response of series and parallel RLC circuits.
- Classify the response of second-order circuits.

Lecture Outline

- Characteristics of Second-Order Circuits
- Undriven Series RLC Circuit
- Characteristic Equation and its Roots
- Types of Responses
 - Overdamped
 - Critically-Damped
 - Underdamped
- Undriven Parallel RLC Circuit
- Examples

Second-Order Circuits

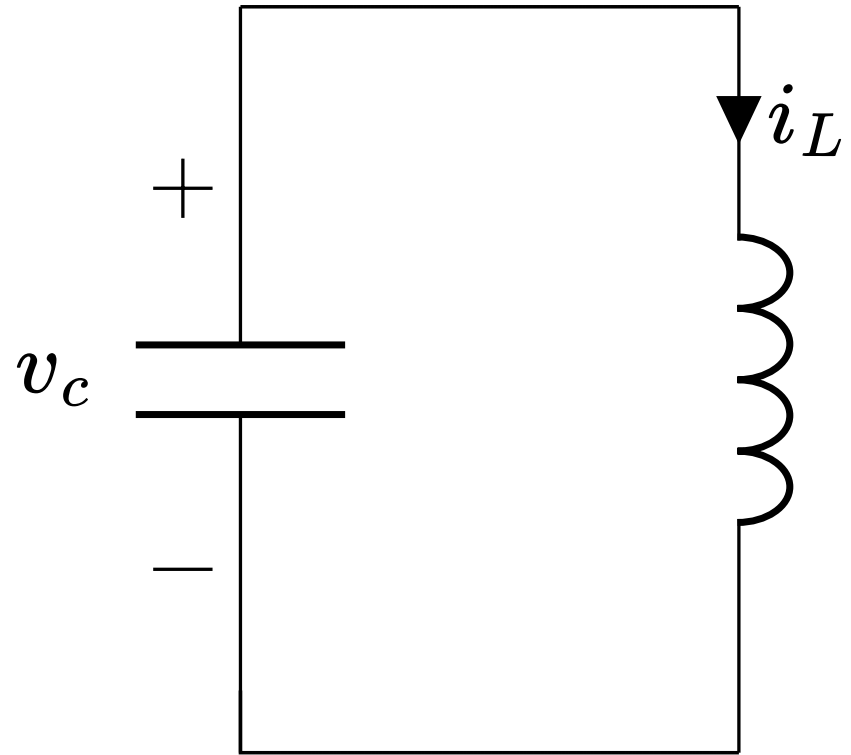
- Any circuit containing **two energy storage elements** with **independent** states.



- Examples: clock pendulums, automobile suspensions, tuned filters in radios, inter-chip digital interconnections.
- Second-order circuits are described by **second-order differential equations**.

Undriven LC Circuit

- It is
 - the simplest second-order circuit.
 - lossless.
 - undriven because it has no external stimuli.
- It allows us to focus on internal/homogenous behavior
- Of course, something energized the circuit before it becomes undriven.



Undriven LC Circuit

Performing nodal analysis:

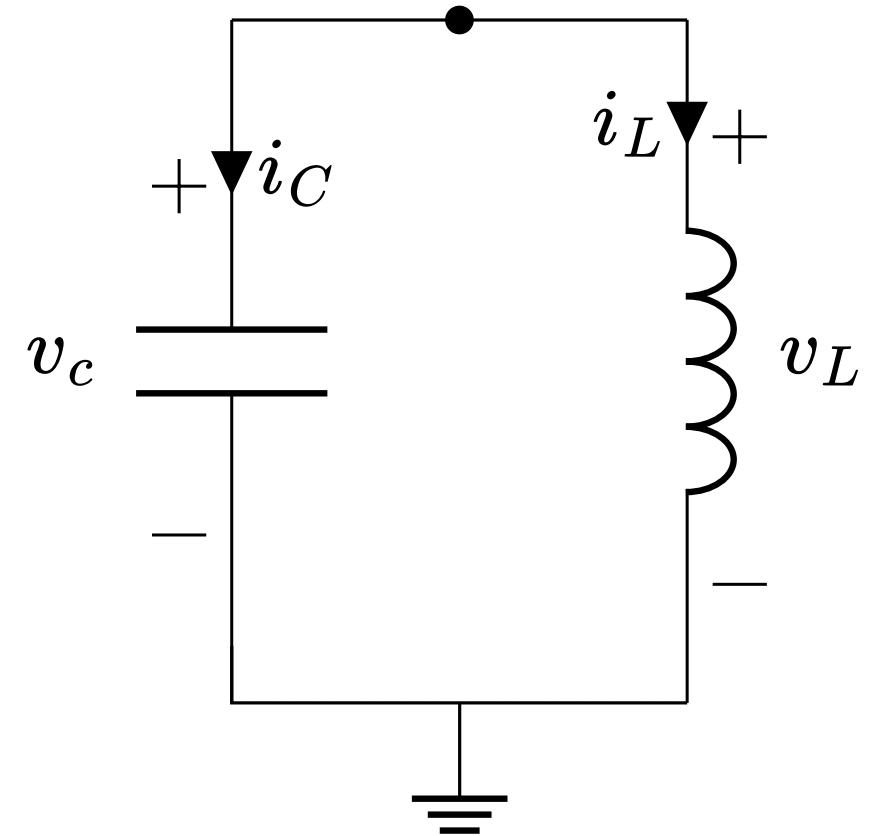
$$v_C(t) = v_L(t) = v(t)$$

Note that:

$$v(t) = L \frac{di_L(t)}{dt} \Rightarrow i_L(t) = \frac{1}{L} \int_{-\infty}^t v(t)$$

From KCL:

$$i_C(t) + i_L(t) = 0$$
$$C \frac{dv(t)}{dt} + \frac{1}{L} \int_{-\infty}^t v(t) = 0$$



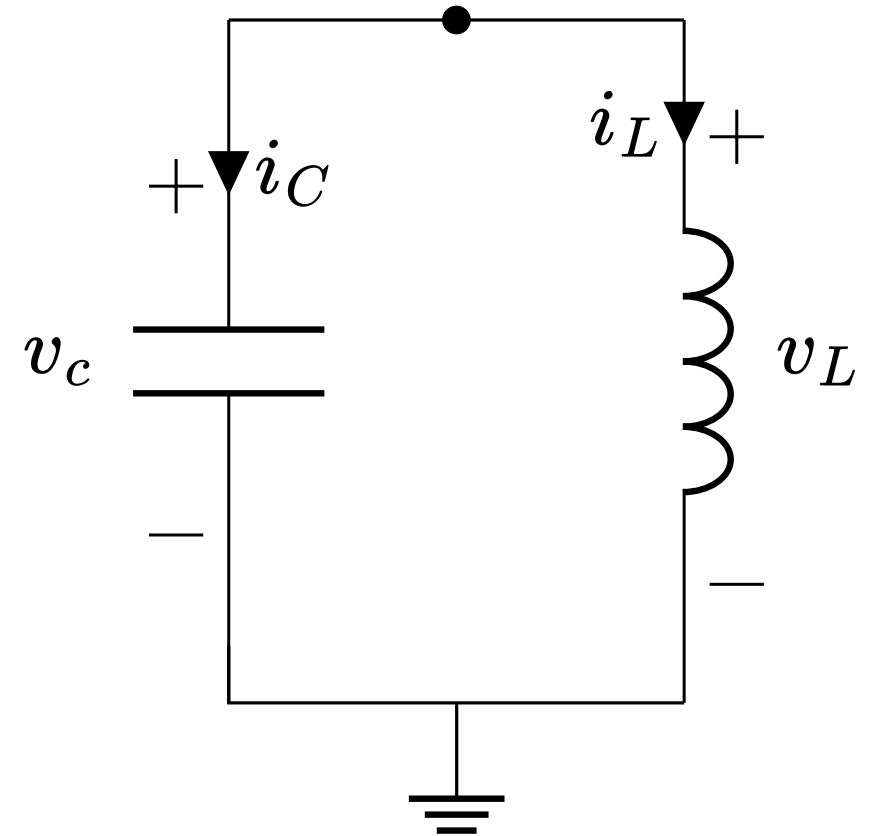
Undriven LC Circuit

$$C \frac{dv(t)}{dt} + \frac{1}{L} \int_{-\infty}^t v(t) dt = 0$$

Differentiating once with respect to t and dividing by C :

$$\frac{d^2v(t)}{dt^2} + \frac{1}{LC} v(t) = 0$$

Now we solve for $v(t)$.



Recall: Complete Solution of Differential Equations

The complete solution $x(t)$ of any differential equation can be decomposed into the *homogeneous* and *particular* solutions, i.e.,

$$x(t) = x_h(t) + x_p(t)$$

Therefore, for any second-order system, we also have a homogeneous response and a steady-state response.

Recall: Natural Response of First-Order Systems

The **natural response** for first-order circuits is in exponential form:

$$x_h(t) = k e^{-\frac{t}{\tau}}$$

The homogeneous solution is determined by the initial conditions of the system, without the influence of any external forces.

*Maybe we can **assume** that the solution for second-order circuits takes a similar form?*

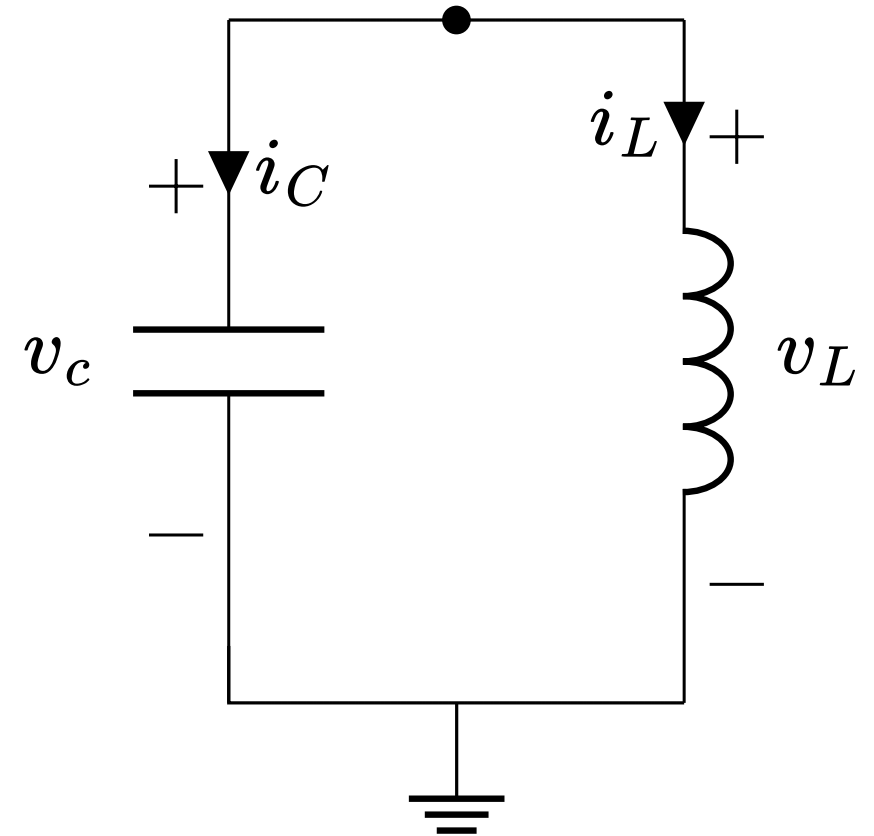
Undriven LC Circuit

Assume that the solution takes the exponential form:

$$v(t) = Ae^{st}$$

Substituting:

$$\begin{aligned}\frac{d^2 v(t)}{dt^2} + \frac{1}{LC} v(t) &= 0 \\ \frac{d^2}{dt^2} (Ae^{st}) + \frac{1}{LC} (Ae^{st}) &= 0 \\ As^2 e^{st} + A \frac{1}{LC} e^{st} &= 0\end{aligned}$$

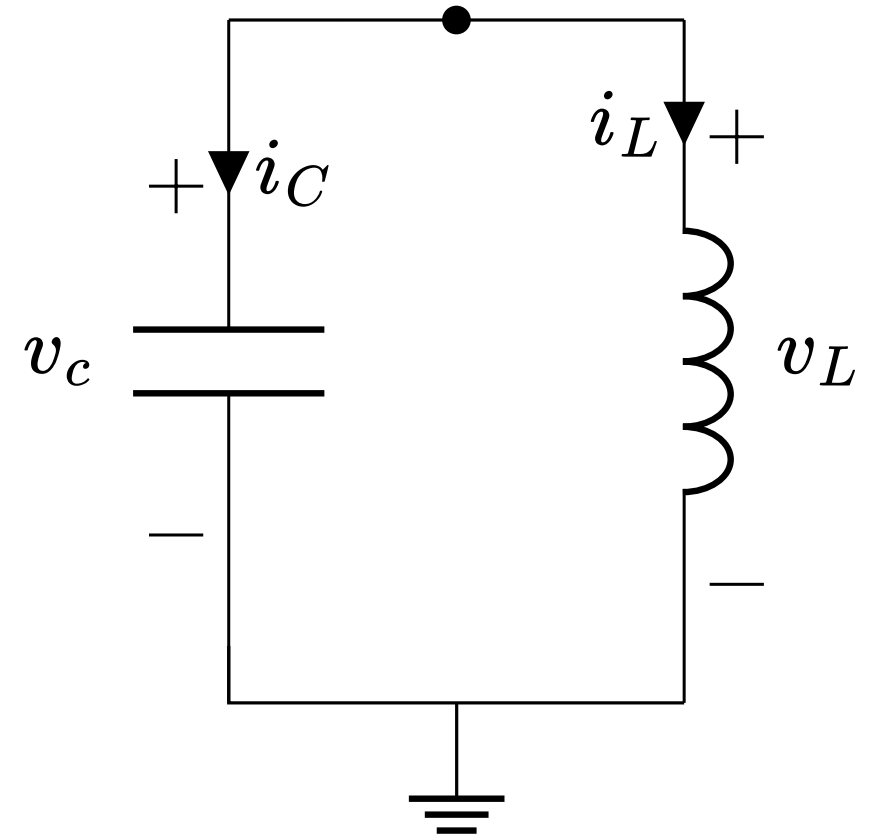


Undriven LC Circuit

$$As^2e^{st} + A\frac{1}{LC}e^{st} = 0$$
$$Ae^{st}\left(s^2 + \frac{1}{LC}\right) = 0$$

For the equation to hold true, either

- $A = 0$ (implies that $v(t) = 0$ **impossible!**)
- $s^2 + \frac{1}{LC} = 0$ [our **characteristic equation!**]
 - Solutions are $s_{1,2} = \pm j\omega_0$ where $\omega_0 = \sqrt{\frac{1}{LC}}$. [a.k.a. **natural frequency**]



Undriven LC Circuit

Then,

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$v(t) = A_1 e^{j\omega_0 t} + A_2 e^{-j\omega_0 t}$$

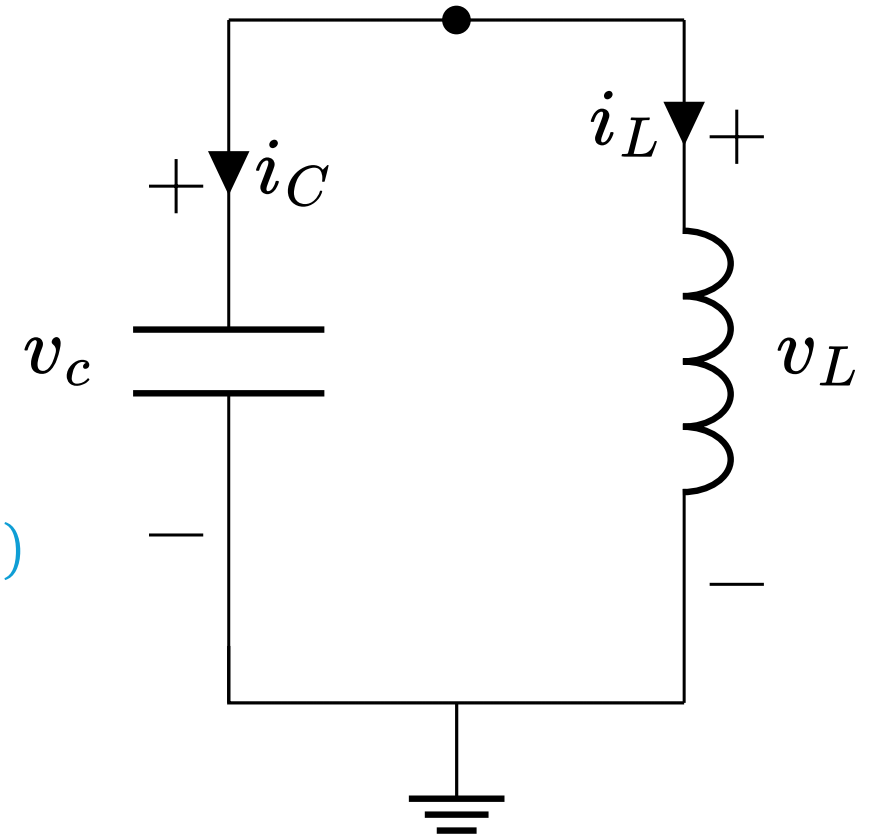
Recall Euler's identity: $e^{\pm j\theta} = \cos(\theta) \pm j\sin(\theta)$

$$v(t) = A_1 (\cos(\omega_0 t) + j\sin(\omega_0 t)) + A_2 (\cos(\omega_0 t) - j\sin(\omega_0 t))$$

$$v(t) = (A_1 + A_2) \cos(\omega_0 t) + j(A_1 - A_2) \sin(\omega_0 t)$$

$$v(t) = K_1 \cos(\omega_0 t) + K_2 \sin(\omega_0 t)$$

Note: K_1 and K_2 are still unknown constants. These can be determined based on system's initial conditions.



Undriven LC Circuit

$$v(t) = K_1 \cos(\omega_0 t) + K_2 \sin(\omega_0 t)$$

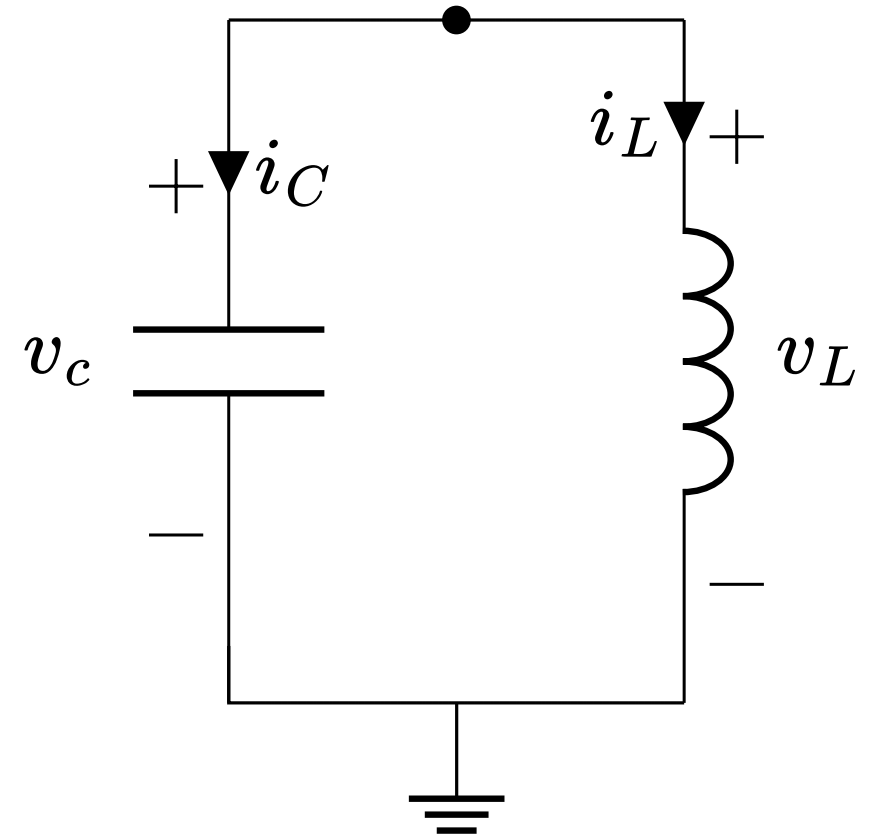
Note that the initial conditions are:

- $v(0) = v_C(0)$
- $\frac{dv(0)}{dt} = \frac{i_C}{C} = \frac{-i_L(0)}{C}$

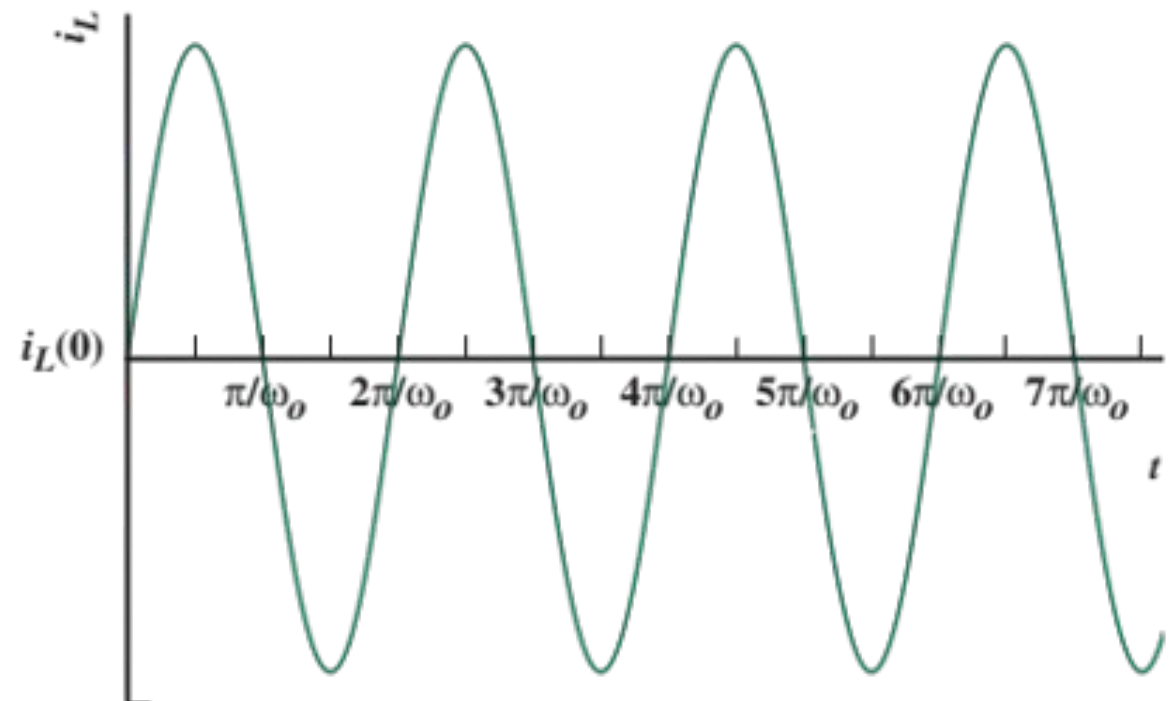
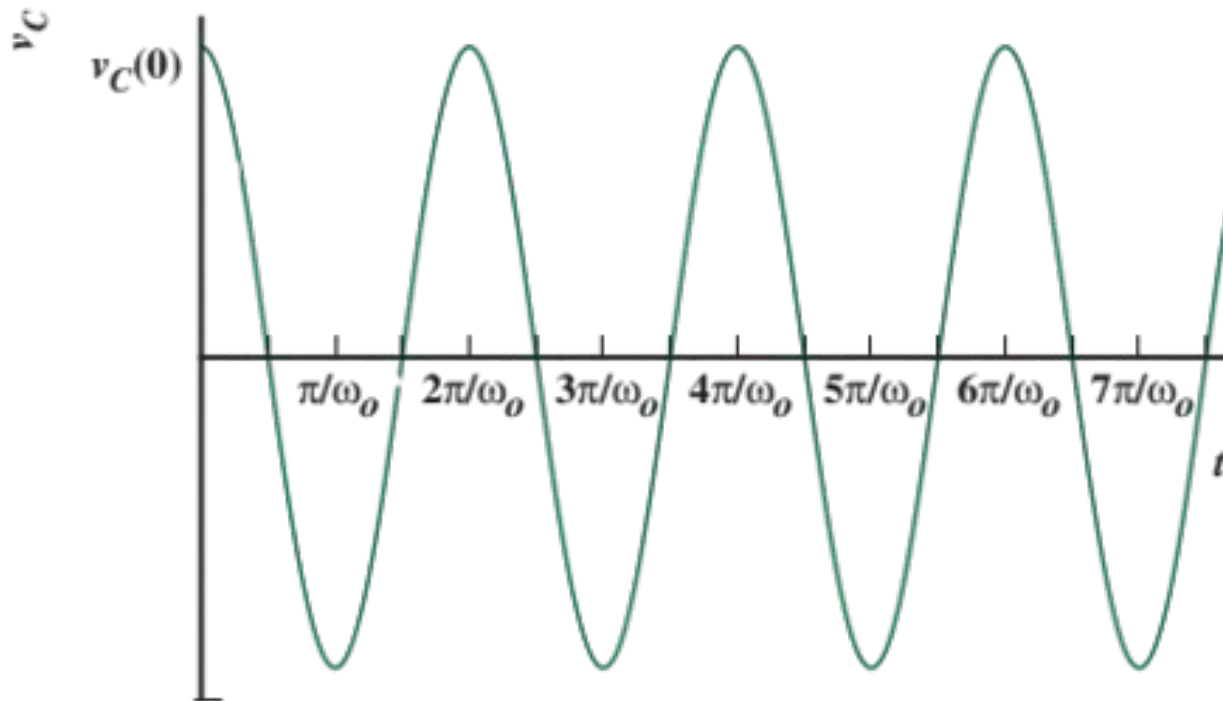
Then,

$$\begin{aligned} v(0) &= K_1 \cos(0) + K_2 \sin(0) \Rightarrow K_1 = v_C(0) \\ \frac{dv(0)}{dt} &= -K_1 \omega_0 \sin(0) + K_2 \omega_0 \cos(0) \Rightarrow K_2 = -\frac{1}{\omega_0 C} i_L(0) \end{aligned}$$

$$\therefore v(t) = v_C(0) \cos(\omega_0 t) - \sqrt{\frac{L}{C}} i_L(0) \sin(\omega_0 t)$$



Undriven LC Circuit

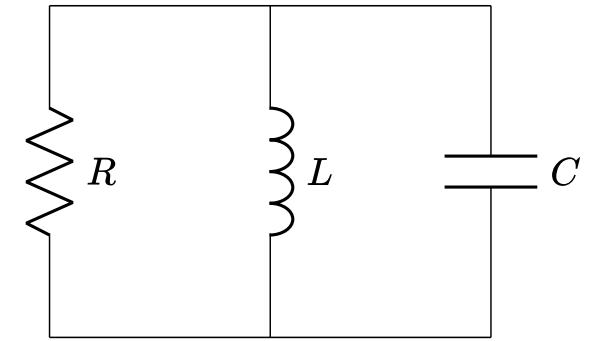


Undriven LC Circuit: Remarks

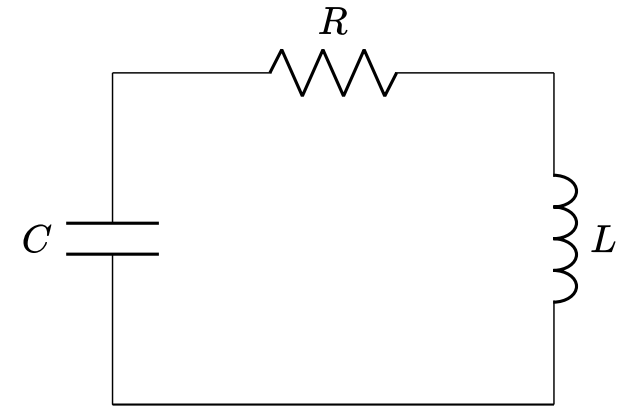
- The capacitor voltage and inductor current are both sinusoidal functions.
- The frequency is the magnitude of the natural frequency.
- The capacitor voltage leads the inductor current by 90 degrees.

Undriven RLC Circuits

- The **presence of resistor** is *naturally occurring*
 - e.g. due to the capacitors dielectric leakage or the inductors winding resistance
- Resistors are intentionally added to **make oscillations decay**
- Typical models encountered are **parallel** and **series** RLC circuits



Parallel RLC



Series RLC

Undriven Series RLC Circuit

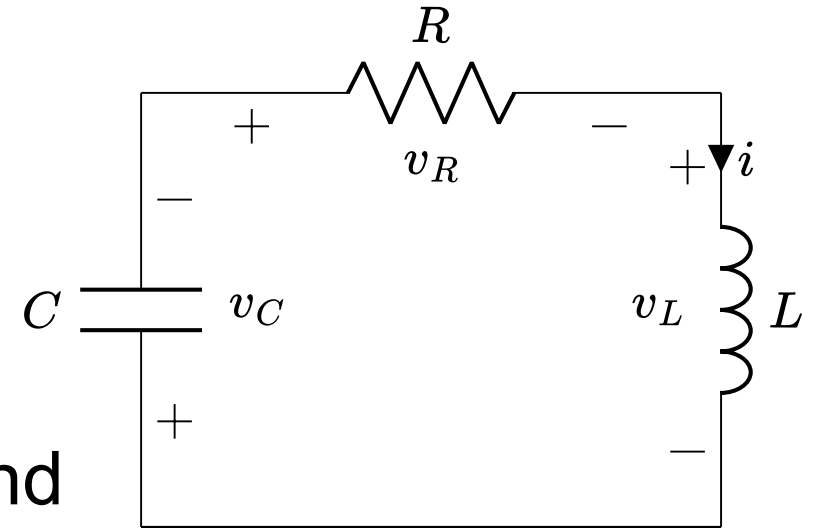
Derivation of the Characteristic Equation

KVL on the single loop yields:

$$v_C + v_R + v_L = 0$$
$$\frac{1}{C} \int_{-\infty}^t i(t) dt + Ri(t) + L \frac{di(t)}{dt} = 0$$

Differentiating both sides with respect to t and dividing by L :

$$\frac{1}{LC} i(t) + \frac{R}{L} \frac{di(t)}{dt} + \frac{d^2 i(t)}{dt^2} = 0$$



Undriven Series RLC Circuit

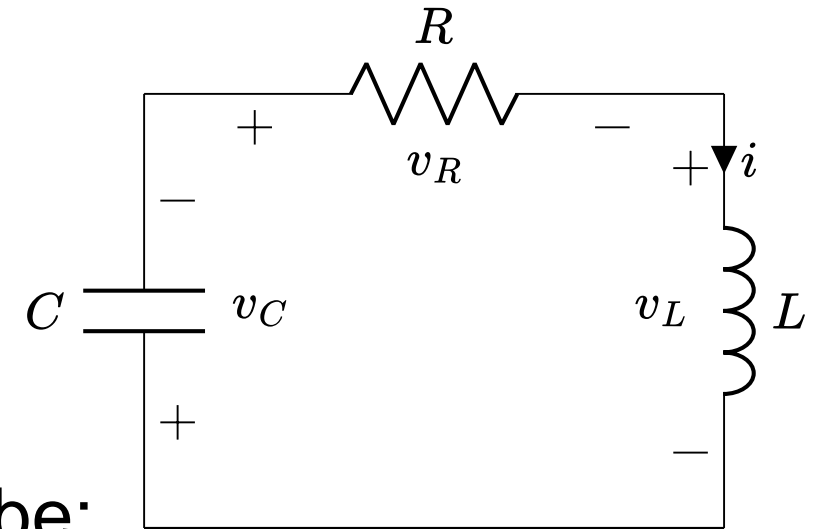
Derivation of the Characteristic Equation

Rearranging terms:

$$\frac{d^2 i(t)}{dt^2} + \frac{R}{L} \frac{di(t)}{dt} + \frac{1}{LC} i(t) = 0$$

We can solve for the homogeneous solution $i(t)$ by assuming the form of the solution to be:

$$i(t) = Ae^{st}$$



Undriven Series RLC Circuit

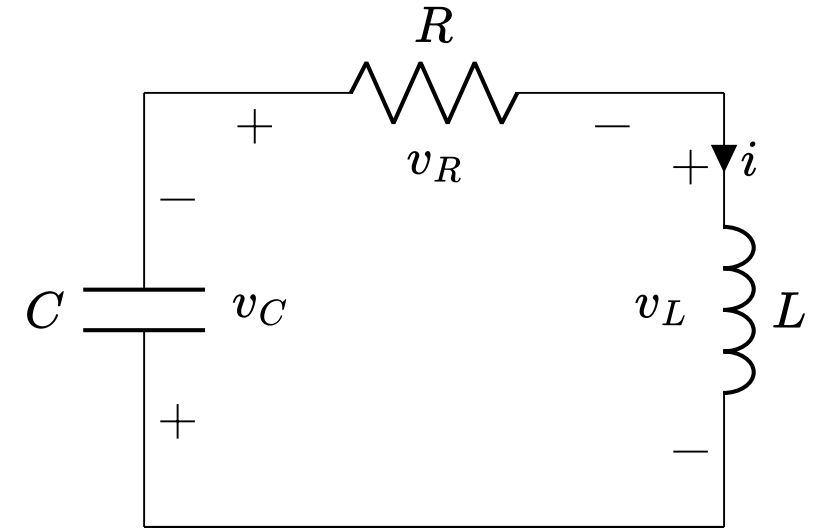
Derivation of the Characteristic Equation

Substituting the assumed solution in our derived equation,

$$\frac{d^2[Ae^{st}]}{dt} + \frac{R}{L} \frac{[Ae^{st}]}{dt} + \frac{1}{LC} [Ae^{st}] = 0$$

$$As^2e^{st} + \frac{R}{L} (Ase^{st}) + \frac{Ae^{st}}{LC} = 0$$

$$Ae^{st} \left(s^2 + \frac{R}{L}s + \frac{1}{LC} \right) = 0 \longrightarrow$$



$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

Characteristic Equation

Undriven Series RLC Circuit

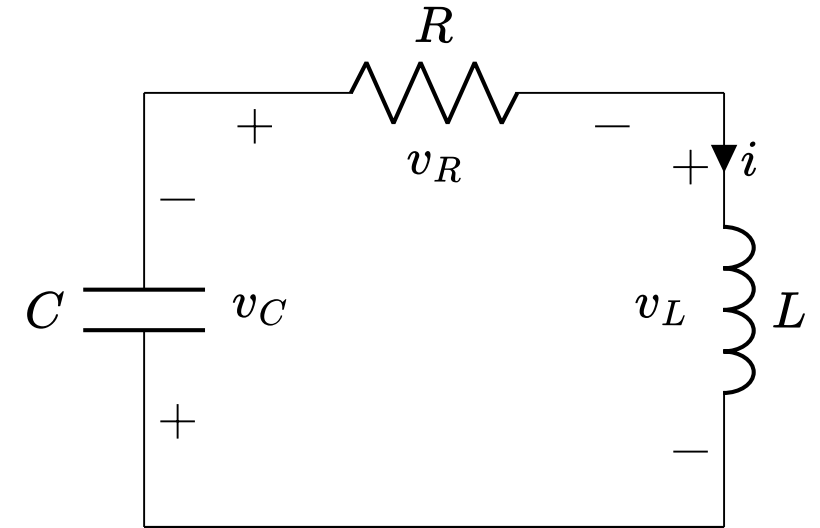
Derivation of the Characteristic Equation

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

We can solve for s using the quadratic equation,

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

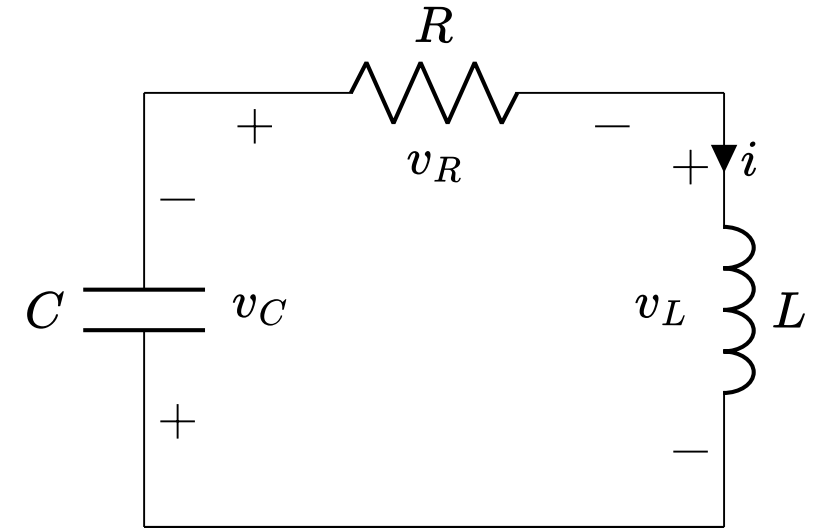
Given $as^2 + bs + c = 0$



Undriven Series RLC Circuit

Derivation of the Characteristic Equation

$$s_1, s_2 = \frac{-\left(\frac{R}{L}\right) \pm \sqrt{\left(\frac{R}{L}\right)^2 - 4(1)\left(\frac{1}{LC}\right)}}{2(1)}$$
$$= -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \left(\frac{1}{LC}\right)}$$



Undriven Series RLC Circuit

Derivation of the Characteristic Equation

$$s_1, s_2 = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \left(\frac{1}{LC}\right)}$$

Key terms:

- $\alpha = \frac{R}{2L} \rightarrow$ damping factor (in nepers per second or $\frac{Np}{s}$)
- $\omega_0 = \frac{1}{\sqrt{LC}} \rightarrow$ undamped natural frequency (in $\frac{rad}{s}$)
 - also known as undamped resonant frequency
- $s_1, s_2 \rightarrow$ natural frequencies (in $\frac{Np}{s}$)

Undriven Series RLC Circuit

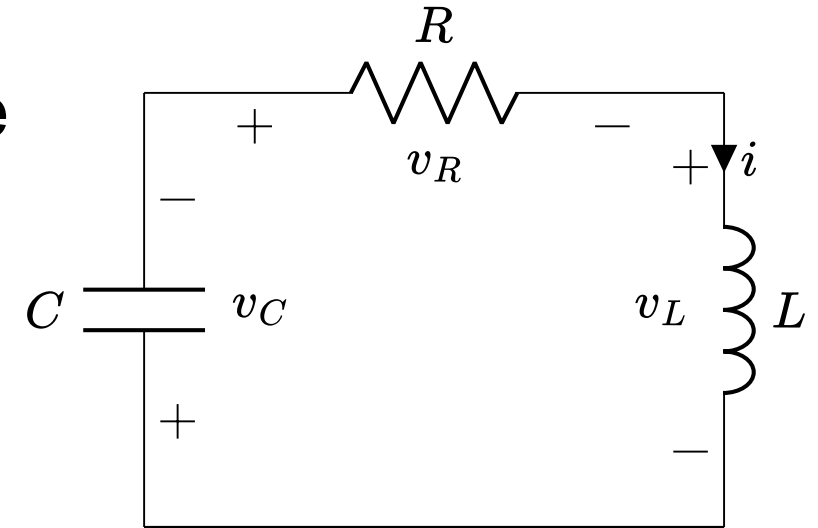
Derivation of the Characteristic Equation

The characteristic equation can therefore be rewritten in the following standard form:

$$s^2 + 2\alpha s + \omega_0^2 = 0$$

And the solutions to the characteristic equation are:

$$s_1, s_2 = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$



- CE: $s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$
- SOLN: $s_1, s_2 = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \left(\frac{1}{LC}\right)}$

Undriven Series RLC Circuit

Derivation of the Characteristic Equation

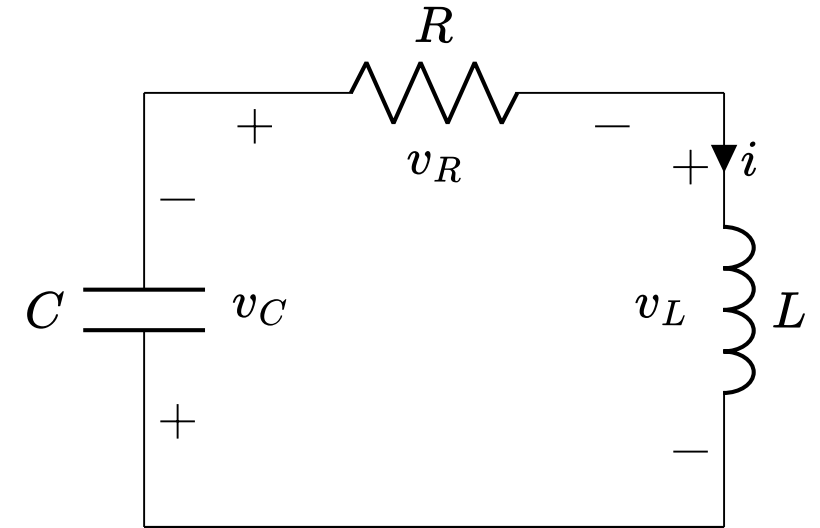
Then

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$i(t) = A_1 e^{\left(-\alpha + \sqrt{\alpha^2 - \omega_0^2}\right)t} + A_2 e^{\left(-\alpha - \sqrt{\alpha^2 - \omega_0^2}\right)t}$$

$$i(t) = e^{-\alpha t} \left(A_1 e^{\left(\sqrt{\alpha^2 - \omega_0^2}\right)t} + A_2 e^{\left(-\sqrt{\alpha^2 - \omega_0^2}\right)t} \right)$$

Note: A_1 and A_2 are still unknown constants. These can be determined based on system's initial conditions.



Types of Responses

$$x(t) = e^{-\alpha t} \left(A_1 e^{\left(\sqrt{\alpha^2 - \omega_0^2} \right) t} + A_2 e^{\left(-\sqrt{\alpha^2 - \omega_0^2} \right) t} \right)$$

The response of v_C and i_L depends on the relation of α and ω_0 .

- $\alpha > \omega_0$: **overdamped response**
- $\alpha = \omega_0$: **critically-damped response**
- $\alpha < \omega_0$: **underdamped response**

Overdamped Case

$$s_1, s_2 = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Since $\alpha > \omega_0$, we have **two real and distinct roots**, s_1 and s_2 .
Hence,

$$x(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Overdamped Case

- When R becomes very large, the roots become the reciprocal of the time constants

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_o^2} = -\alpha \left(1 + \sqrt{1 - \left(\frac{\omega_o}{\alpha} \right)^2} \right) \approx -2\alpha = -\frac{R}{L}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_o^2} = -\alpha \left(1 - \sqrt{1 - \left(\frac{\omega_o}{\alpha} \right)^2} \right) \approx -\alpha \frac{\omega_o^2}{2\alpha^2} = -\frac{1}{RC}$$

Critically-damped Case

$$s_1, s_2 = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Since $\alpha = \omega_0$, we have **two real and equal roots**, and $e^{s_1 t}$ and $e^{s_2 t}$ are no longer independent functions

Hence,

$$x(t) = (A_1 t + A_2)e^{st}$$

Underdamped Case

$$s_1, s_2 = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Since $\alpha < \omega_0$, the value for $\alpha^2 - \omega_0^2$ becomes negative, resulting to **two complex roots**.

$$s_1, s_2 = -\alpha \pm \sqrt{-(\omega_0^2 - \alpha^2)}$$

Note that $j = \sqrt{-1}$. Let $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

$$s_1, s_2 = -\alpha \pm j\omega_d$$

Underdamped Case

$$x(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

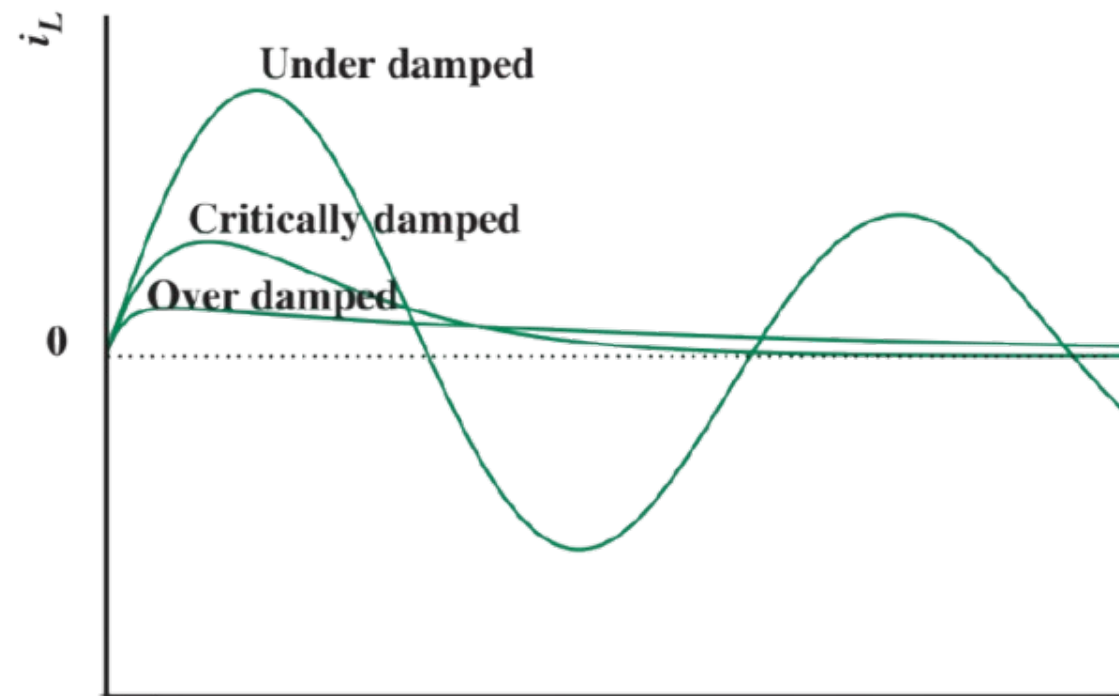
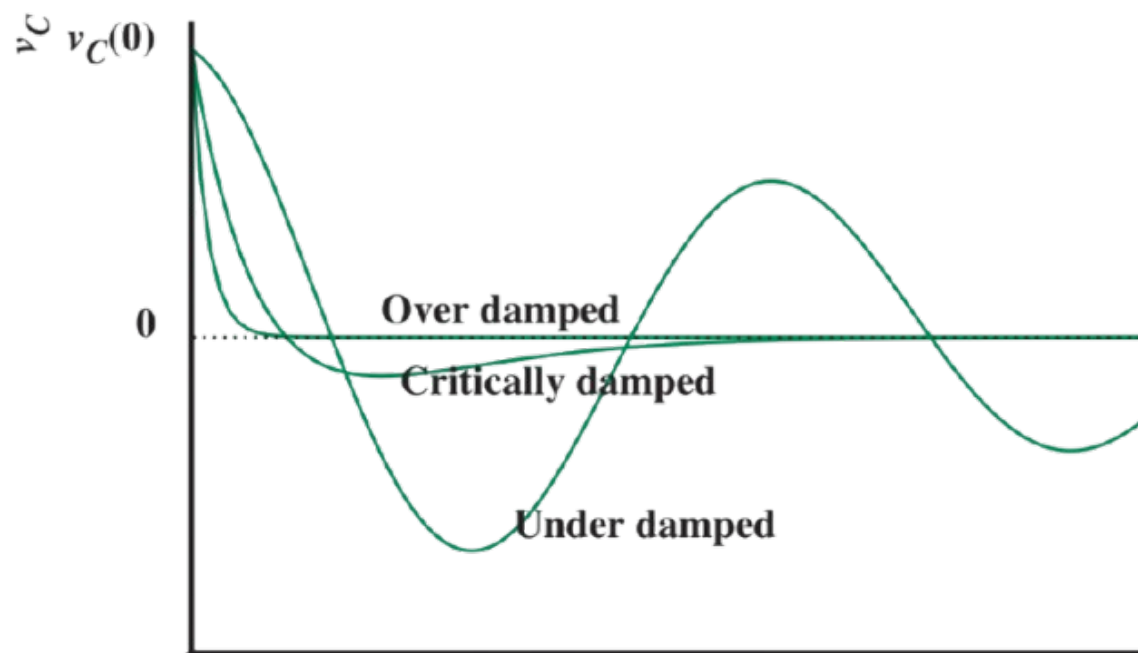
$$x(t) = e^{-\alpha t} (A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t})$$

Noting that $e^{j\omega_d t} = \cos(\omega_d t) + j \sin(\omega_d t)$

$$x(t) = e^{-\alpha t} [(A_1 + A_2) \cos(\omega_d t) + j(A_1 - A_2) \sin(\omega_d t)]$$

$$x(t) = e^{-\alpha t} [B_1 \cos(\omega_d t) + B_2 \sin(\omega_d t)]$$

Types of Responses



Undriven Parallel RLC Circuit

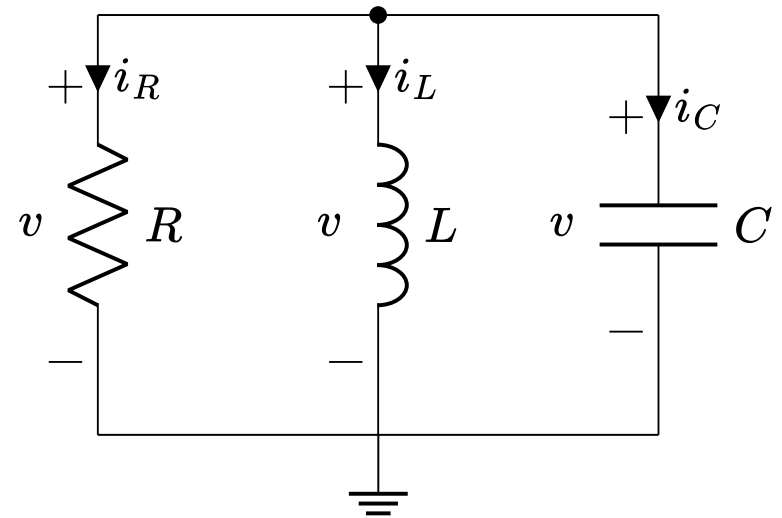
Derivation of the Characteristic Equation

KCL on the top node yields:

$$\frac{v(t)}{R} + \frac{1}{L} \int_{-\infty}^t v(t) dt + C \frac{dv(t)}{dt} = 0$$

Differentiating both sides with respect to t and dividing by C :

$$\frac{1}{RC} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) + \frac{d^2 v(t)}{dt^2} = 0$$



Undriven Parallel RLC Circuit

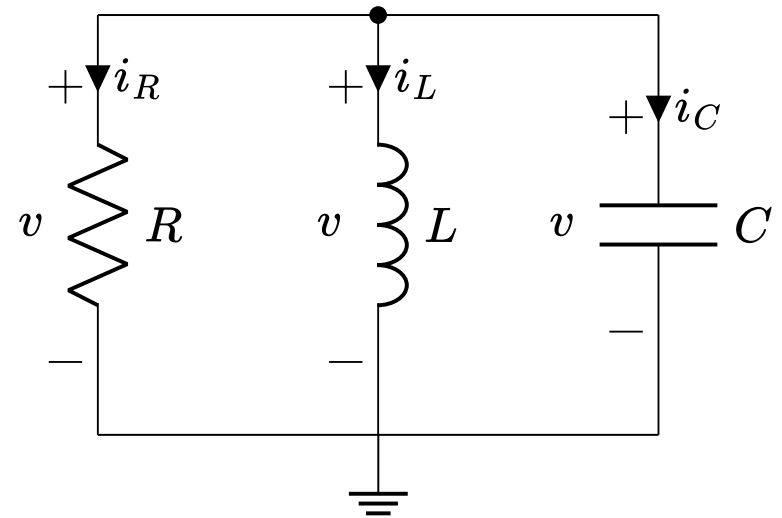
Derivation of the Characteristic Equation

Rearranging terms:

$$\frac{d^2 v_C(t)}{dt^2} + \frac{1}{RC} \frac{dv_C(t)}{dt} + \frac{1}{LC} v_C(t) = 0$$

We can solve for the homogeneous solution $v(t)$ by assuming the form of the solution to be:

$$v(t) = Ae^{st}$$



Undriven Parallel RLC Circuit

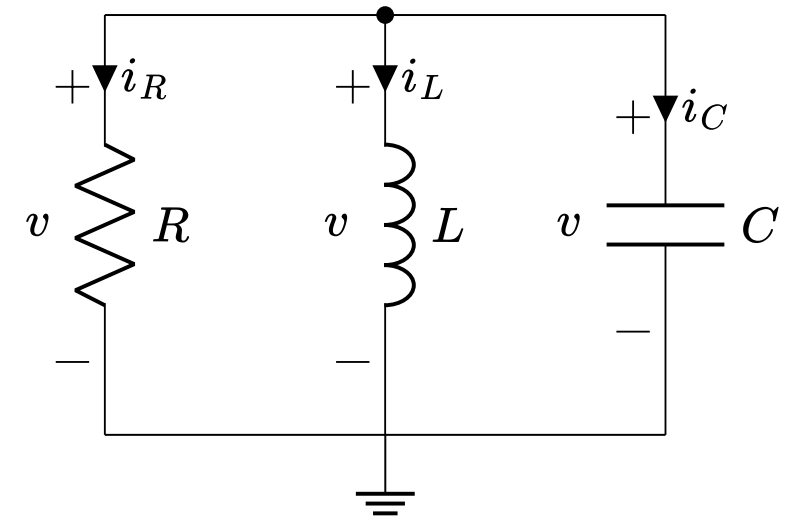
Derivation of the Characteristic Equation

Substitution results to the following DE:

$$Ae^{st} \left(s^2 + \frac{1}{RC} s + \frac{1}{LC} \right) = 0$$

Where the characteristic equation is:

$$s^2 + \frac{1}{RC} s + \frac{1}{LC} = 0$$



Undriven Parallel RLC Circuit

Derivation of the Characteristic Equation

Considering the standard form of the characteristic equation:

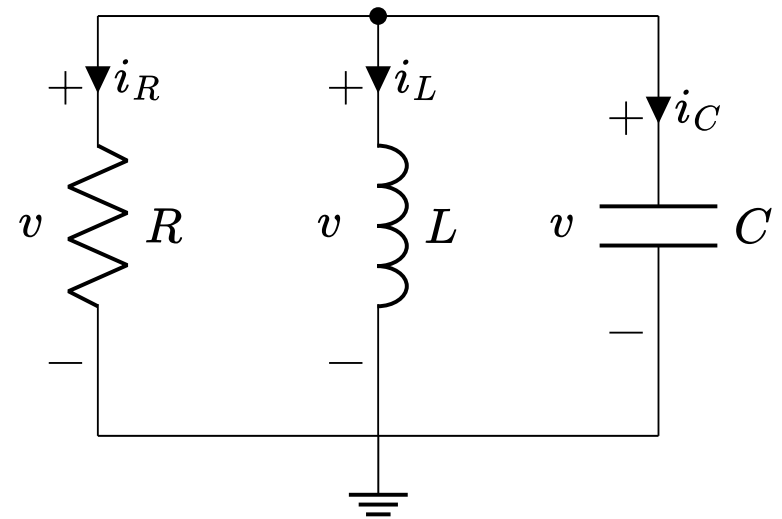
$$s^2 + 2\alpha s + \omega_0^2 = 0$$

Then for parallel RLC circuits:

- $\alpha = \frac{1}{2RC}$

- $\omega_0 = \frac{1}{\sqrt{LC}}$

- $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$



Undriven Parallel RLC Circuit

The response of parallel RLC circuits are also categorized into the following:

- **Overdamped ($\alpha > \omega_0$)**

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

- **Critically damped ($\alpha = \omega_0$)**

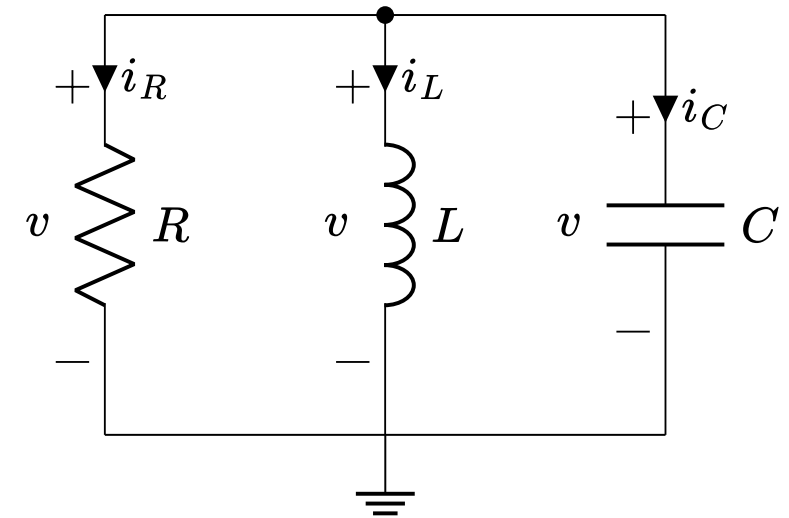
$$v(t) = (A_1 t + A_2) e^{-\alpha t}$$

- **Underdamped ($\alpha < \omega_0$)**

$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

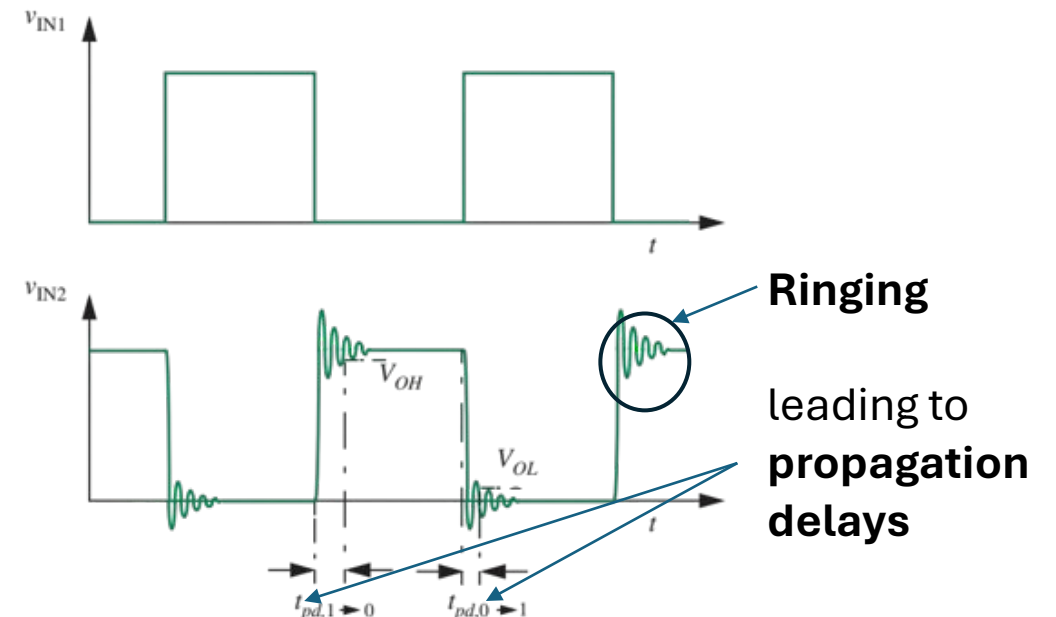
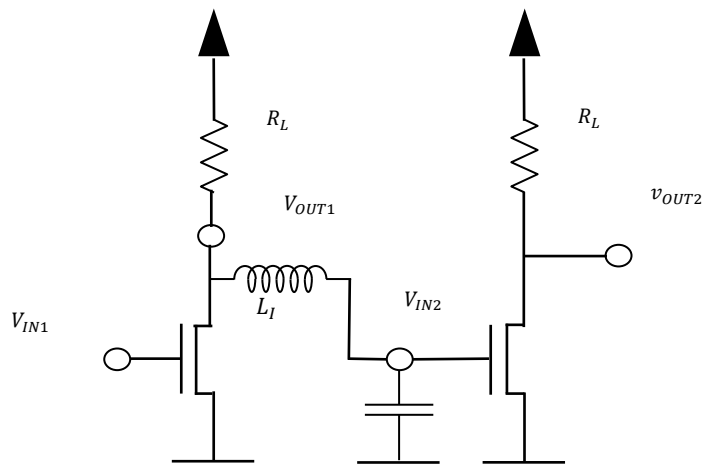
$$s_{1,2} = -\alpha \pm j\omega_d$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$



More on the Underdamped Response

The oscillatory response for the underdamped can be thought of as the energy storage elements juggling their energies back-and-forth. The damped oscillation is what we call *ringing*, which is often exhibited in poorly designed amplifiers, or semiconductor devices with a lot of parasitic inductances and capacitances. Ringing increases the rise time or fall time of the output waveform (more often called as *propagation delay*).



Circuit Analysis for the Natural Response of RLC Circuits

1. **Redraw the circuit** by checking the conditions at the following:
 - $x(0^-)$: a long time before the switching.
 - $x(0^+)$: immediately after switching (recall the continuity properties)
2. **Solve for the initial conditions** for the capacitor $(v_C(0^+), \frac{dv_C(0^+)}{dt})$ and inductor $(i_L(0^+), \frac{di_L(0^+)}{dt})$.
3. Determine if the resulting circuit can be simplified into series or parallel RLC. Then, **find α and ω_0** .
Note that $\alpha_{series} = \frac{R}{2L}$ and $\alpha_{parallel} = \frac{1}{2RC}$
4. Determine the **type of response**:
 - Overdamped ($\alpha > \omega_0$): $x(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$
 - Critically damped ($\alpha = \omega_0$): $x(t) = (A_1 t + A_2) e^{-\alpha t}$
 - Underdamped ($\alpha < \omega_0$): $x(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$
5. **Evaluate the unknown coefficients** using the initial conditions of the system.
6. Substitute all values into the determined response type.

Summary of Relevant Equations for Undriven RLC Circuits

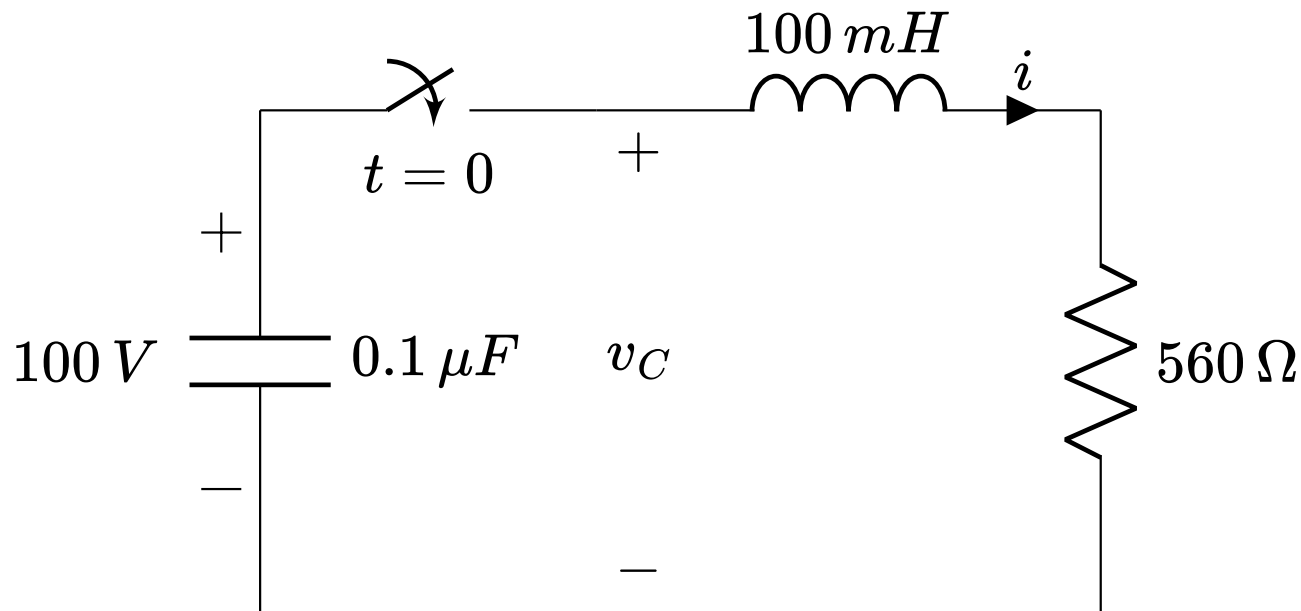
SYSTEM	TRANSIENT RESPONSE	INITIAL CONDITIONS
UNDERDAMPED $\alpha < \omega_0$	$f(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$ $\text{where } \omega_d = \sqrt{\omega_0^2 - \alpha^2}$	$f(0) = B_1$ $f'(0) = -\alpha B_1 + \omega_d B_2$
CRITICALLY DAMPED $\alpha = \omega_0$	$f(t) = e^{-\alpha t}(A_1 + A_2 t)$	$f(0) = A_1$ $f'(0) = -\alpha A_1 + A_2$
OVERDAMPED $\alpha > \omega_0$	$f(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ $\text{where } s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$	$f(0) = A_1 + A_2$ $f'(0) = s_1 A_1 + s_2 A_2$

Note:

- Basic Series RLC: $\alpha = \frac{R}{2L}$ and $\omega_0 = \frac{1}{\sqrt{LC}}$
- Basic Parallel RLC: $\alpha = \frac{1}{2RC}$ and $\omega_0 = \frac{1}{\sqrt{LC}}$

Example 1

The capacitor is charged to 100 V and at $t = 0$, the switch closes. Find $i(t)$ for $t \geq 0$.



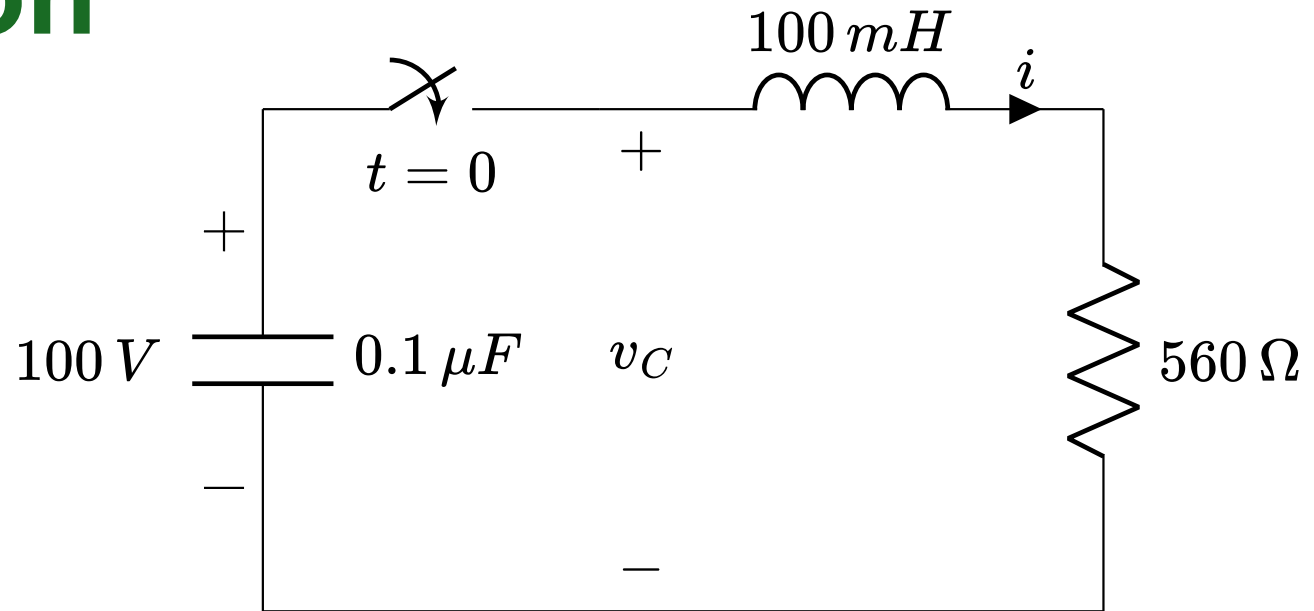
Example 1: Solution

At $t = 0^+$:

- $v_C(0^+) = 100 \text{ V}$
- $i_L(0^+) = 0 \text{ A}$

Meanwhile,

- $\alpha = \frac{R}{2L} = \frac{560}{2(0.1)} = 2800 \text{ rad/s}$
- $\omega_0 = \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{0.1(0.1\mu)}} = 10000 \text{ rad/s}$
- $\alpha < \omega_0 \rightarrow \text{underdamped}$



Example 1: Solution

$$\alpha < \omega_0 \rightarrow \textit{underdamped}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 9600 \text{ rad/s}$$

$$i(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

$$i(t) = B_1 e^{-2800t} \cos 9600t + B_2 e^{-2800t} \sin 9600t \text{ (A)}, t \geq 0$$

Example 1: Solution

- Now we must use the coefficients in the equation to satisfy the initial conditions in the circuit:

$$i(t)|_{t=0} \text{ in the equation} = i(t)|_{t=0} \text{ in the circuit}$$

$$\left. \frac{di(t)}{dt} \right|_{t=0} \text{ in the equation} = \left. \frac{di(t)}{dt} \right|_{t=0} \text{ in the circuit}$$

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$$i(0) = B_1 = i_L(0^+) = 0$$

$$B_1 = 0$$

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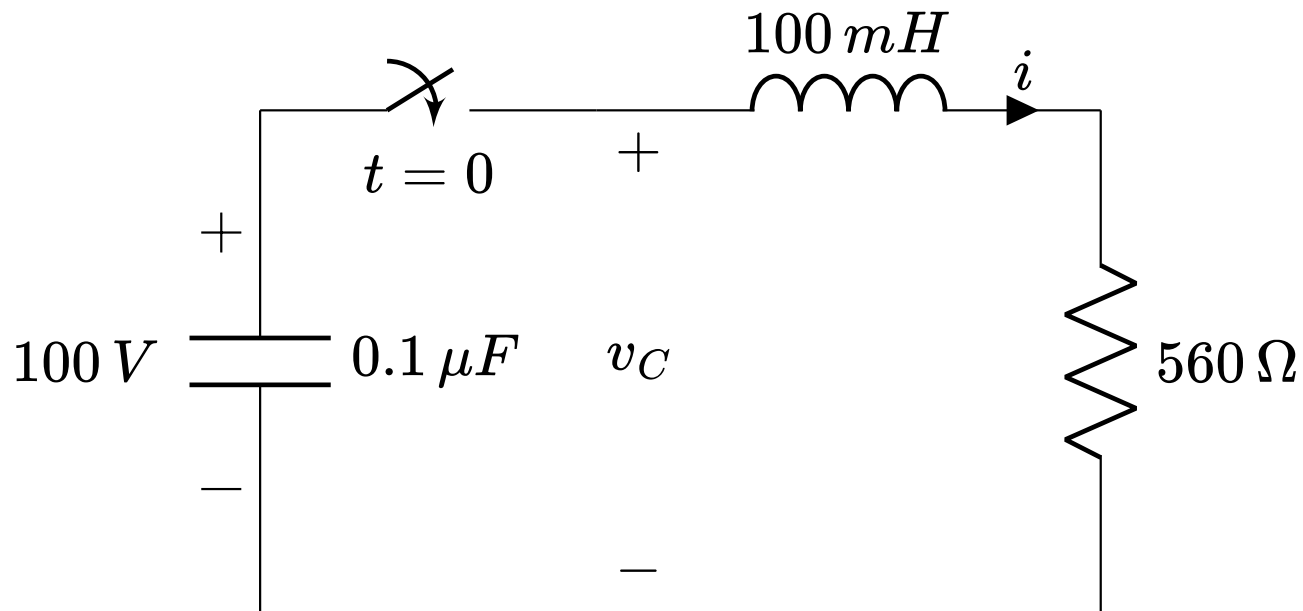
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$$\begin{aligned} i(t) &= B_1 e^{-2800t} \cos 9600t + B_2 e^{-2800t} \sin 9600t \\ \frac{di(0)}{dt} &= -\alpha B_1 + \omega_d B_2 = \frac{di_L(0^+)}{dt} = \frac{v_L}{L} = \frac{100}{0.1} = 1000 \\ -2800(0) + 9600B_2 &= 1000 \\ \mathbf{B_2} &= \mathbf{0.104} \end{aligned}$$

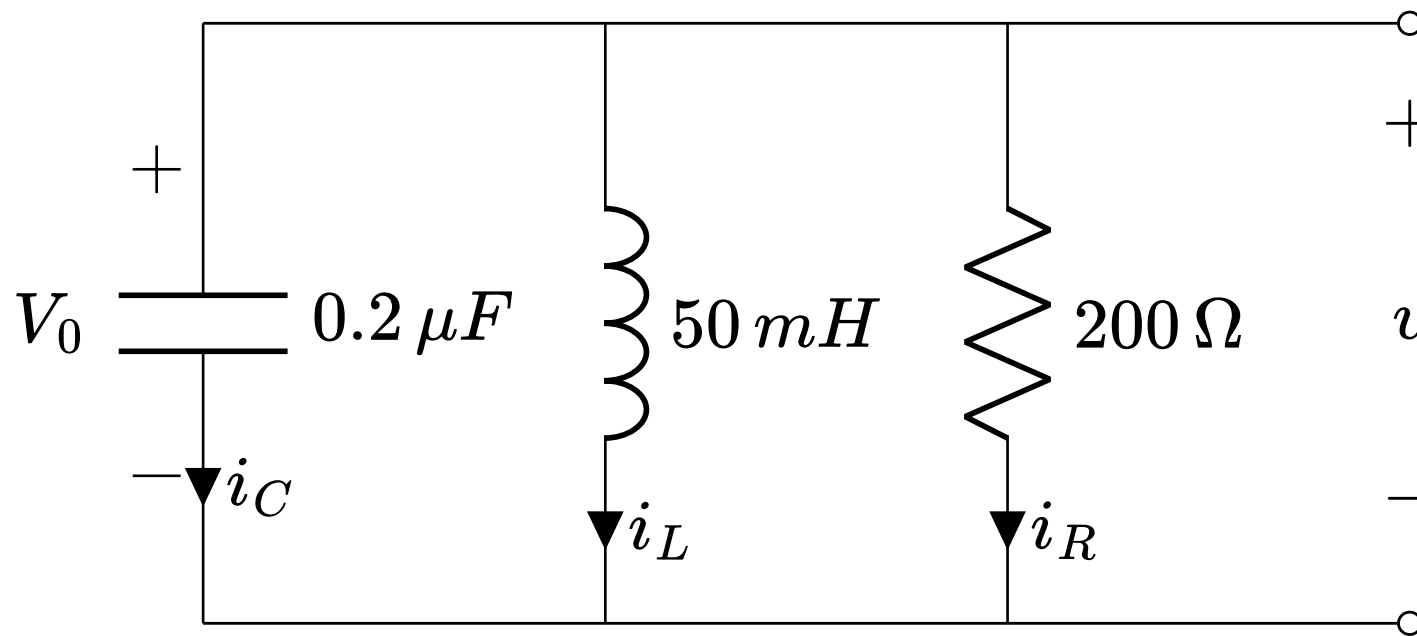
Example 1: Solution

$$\therefore i(t) = 0.104e^{-2800t} \sin 9600t \text{ (A)}, t \geq 0$$



Example 2

Given $V_0 = 12\text{ V}$ at $t = 0^+$ and $I_0 = 30\text{ mA}$, find $v(t)$ for $t \geq 0$.



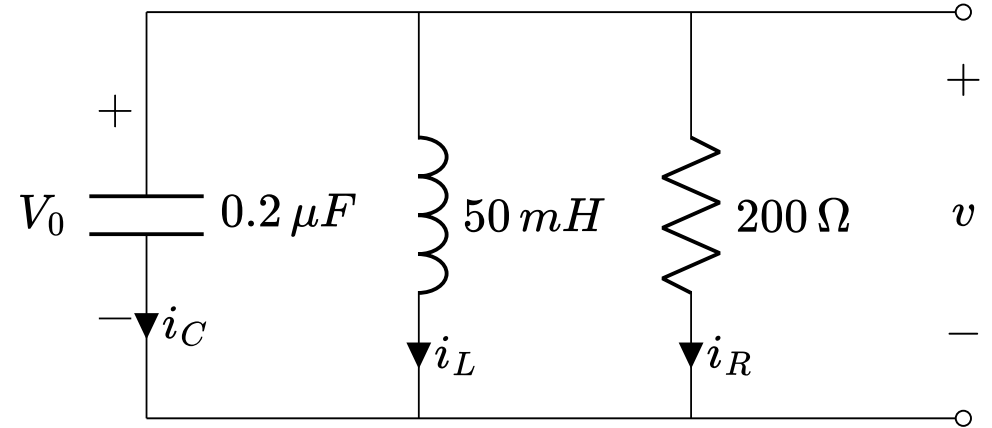
Example 2: Solution

At $t = 0^+$:

- $v_C(0^+) = 12\text{ V}$
- $i_L(0^+) = 30\text{ mA}$

Meanwhile,

- $\alpha = \frac{1}{2RC} = \frac{1}{2(200)(0.2\mu)} = 12500\text{ rad/s}$
- $\omega_0 = \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{0.05(0.2\mu)}} = 10000\frac{\text{rad}}{\text{s}}$
- $\alpha > \omega_0$ (overdamped)



Example 2: Solution

$\alpha > \omega_0$ (overdamped)

$$\begin{aligned}s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -12500 \pm \sqrt{12500^2 - 10000^2} \\ &= -12500 \pm 7500 = -5000 \frac{\text{rad}}{\text{s}}, -20000 \frac{\text{rad}}{\text{s}}\end{aligned}$$

$$\begin{aligned}v(t) &= A_1 e^{s_1 t} + A_2 e^{s_2 t} \\ v(t) &= A_1 e^{-5000t} + A_2 e^{-20000t} \text{ (V), for } t \geq 0\end{aligned}$$

Example 2: Solution

We must use the coefficients in the equations to satisfy the initial conditions in the circuit:

$$v(t)|_{t=0} \text{ in the equation} = v(t)|_{t=0} \text{ in the circuit}$$

$$\left. \frac{dv(t)}{dt} \right|_{t=0} \text{ in the equation} = \left. \frac{dv(t)}{dt} \right|_{t=0} \text{ in the circuit}$$

$$v(t) = A_1 e^{-5000t} + A_2 e^{-20000t}$$
$$v(0) = A_1 e^{-5000(0)} + A_2 e^{-20000(0)} = A_1 + A_2 = V_0 = 12V$$
$$A_1 + A_2 = 12$$

Example 2: Solution

We must use the coefficients in the equations to satisfy the initial conditions in the circuit:

$$v(t)|_{t=0} \text{ in the equation} = v(t)|_{t=0} \text{ in the circuit}$$

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$$v(t) = A_1 e^{-5000t} + A_2 e^{-20000t}$$
$$\frac{dv(0)}{dt} = -5000A_1 e^{-5000(0)} - 20000A_2 e^{-20000(0)} = \frac{i_C(0^+)}{C}$$

Example 2: Solution

We must use the coefficients in the equations to satisfy the initial conditions in the circuit:

$$v(t)|_{t=0} \text{ in the equation} = v(t)|_{t=0} \text{ in the circuit}$$

$$\left. \frac{dv(t)}{dt} \right|_{t=0} \text{ in the equation} = \left. \frac{dv(t)}{dt} \right|_{t=0} \text{ in the circuit}$$

$$\begin{aligned} -5000A_1 - 20000A_2 &= \frac{-i_L(0) - i_R(0)}{C} = \frac{-0.03 - \frac{12}{200}}{0.2\mu} \\ -5000A_1 - 20000A_2 &= -450000 \end{aligned}$$

Example 2: Solution

Solving simultaneously:

$$\begin{aligned} A_1 + A_2 &= 12 \\ -5000A_1 - 20000A_2 &= -450000 \end{aligned}$$

$$\therefore A_1 = -14 \text{ and } A_2 = 26$$

Thus

$$\mathbf{v(t) = -14e^{-5000t} + 26e^{-20000t} \text{ (V) for } t \geq 0}$$

Summary

- A second-order circuit contains two independent energy-storage elements, and its natural response obeys a second-order differential equation.
- The roots of the characteristic equation (the natural frequencies) are set by the damping factor and the undamped natural frequency.
- **Overdamped**: two real, distinct roots; the response decays without oscillating.
- **Critically damped**: two real, equal roots; the response returns to zero in the shortest time without oscillating.
- **Underdamped**: complex conjugate roots; the response oscillates at the damped natural frequency while decaying.
- Solve by the six-step procedure: initial conditions, damping factor and natural frequency, response type, unknown coefficients, complete response.